

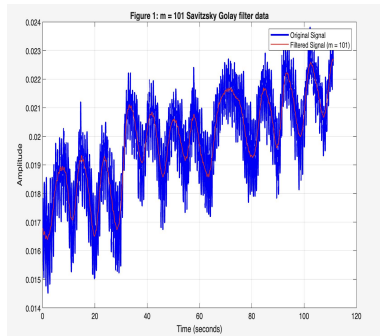
DIGITAL SIGNAL PROCESSING

[Source](#)

Filter Design, Audio Recover, Spectral Analysis, and 2D Frequency-Domain Image Processing

1. Savitzky-Golay Smoothing

Smooth Signal ($M = 101$)

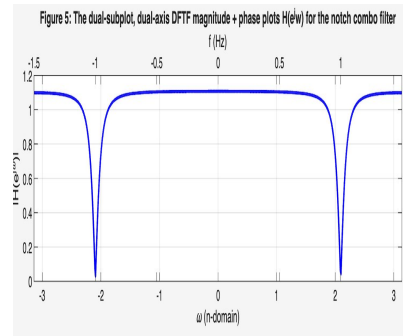


Signal Smoothing with Savitzky-Golay filters

[Code](#)
[Explanation](#)

2. Notch Filter Design

DTFT magnitude for $H(e^{j\omega})$

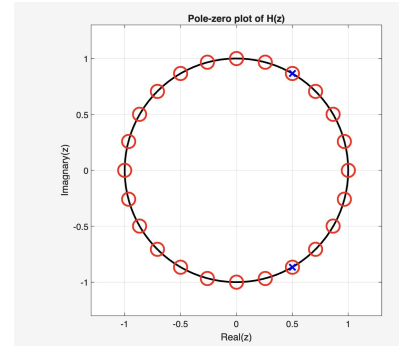


Notch filter magnitude and phase response

[Code](#)
[Explanation](#)

3. Comb + Resonator Bandpass

Pole Zero plot of $H(z)$

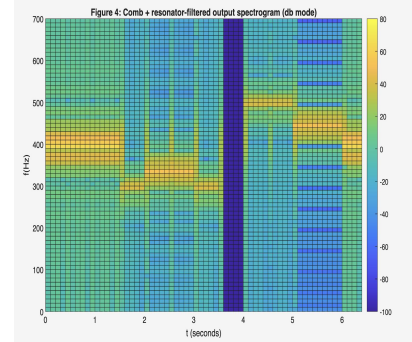


Pole zero plot for comb + resonator filter

[Code](#)
[Explanation](#)

4. Comb + Resonator Audio

Spectrogram of Filtered Output



Recovered audio after Comb + Resonator filter

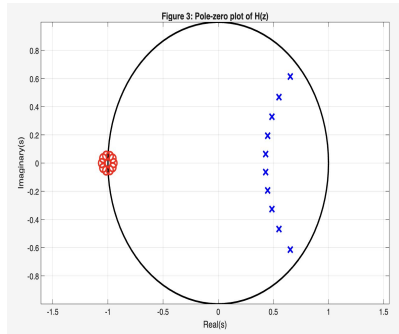
[Code](#)
[Explanation](#)

DIGITAL SIGNAL PROCESSING [Source](#)

Filter Design, Audio Recover, Spectral Analysis, and 2D Frequency-Domain Image Processing

5. Butterworth Basic Design

Pole zero plot of $H(z)$

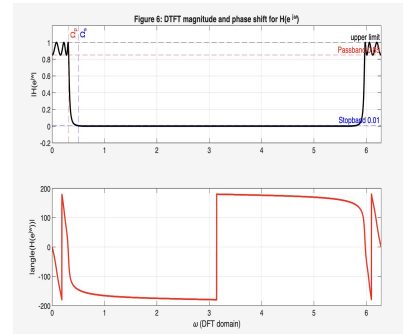


Digital pole zero plot of
Butterworth filter

[Code](#)
[Explanation](#)

6. Chebyshev Basic Design

DTFT magnitude and
phase shift for $H(e^{j\omega})$

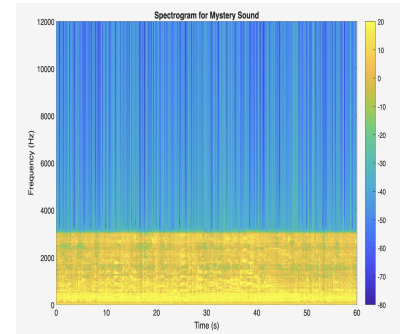


Chebyshev DTFT magnitude
and phase response

[Code](#)
[Explanation](#)

7. Audio Treasure Hunt

Spectrogram for
Mystery Sound

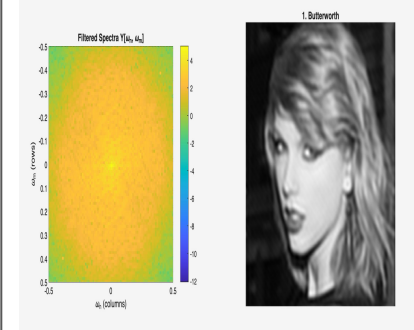


Recovered mystery
audio (Pokemon theme)

[Code](#)
[Explanation](#)

8. Chevy + Butterworth 2D Filter

Butterworth Filter
Output Image



2D Butterworth filtering
in frequency domain

[Code](#)
[Explanation](#)